ArtGuard Final Project Defence Speech

Team 3, SE-2533 · Introduction to Finance · Astana IT University
About 7 minutes of speech + 3:00 video on Slide 5

Slide 1 — Title Assan Dairabaev · ~0:20

Good morning. We are Team 3: Assan Dairabaev, Demezhan Kaidar, Yevgeniy Tyo and Kalybek Zhumabekov, group SE-2533. Our project is ArtGuard. In this final stage we built a local prototype, assessed the risks and tested whether ArtGuard could be an investment. This is an educational simulation: all customers and figures are fictional, and we used only test ETH.

Slide 2 — The licensing problem Assan Dairabaev · ~0:35

Let's start with the problem. Creators need clear evidence of their work and of the terms they agreed to. Buyers need a licence and a reliable delivery process. Co-authors need a traceable record of their shares. Research on blockchain rights management supports these design needs (Ciriello et al., 2023; Frattolillo, 2024). Our first proposed market is visual creators and small- business buyers in Kazakhstan. To be clear: demand and willingness to pay are not yet validated. That is one of our main conclusions.

Slide 3 — Architecture Demezhan Kaidar · ~0:35

Here is what we actually built. The front end is React with a wallet: users fill in forms and sign transactions. The back end is Express with SQLite: it stores accounts, price quotes and protected files. The blockchain layer is a local Anvil EVM: it stores hashes, holds escrow and issues licences to wallets. Files never go on-chain. One honest limitation: the local operator controls Anvil, and the built-in demo wallets use a restricted custodial signer on the server.

Slide 4 — Escrow Demezhan Kaidar · ~0:40

The key design idea is that escrow separates payment from acceptance. First, the creator registers fingerprints of the file and the terms. Second, the buyer locks the exact quoted amount. Third, the seller confirms delivery within 72 hours. Fourth, the buyer has 24 hours to accept or dispute. Fifth, acceptance starts the licence and credits each party's share. Sixth, each recipient withdraws in a separate transaction. A dispute blocks the normal release, and if it stays unresolved, a refund becomes possible after seven days. The timers only make transactions possible; nothing moves automatically.

Slide 5 — Demo video Demezhan Kaidar · ~0:15 + 3:00 video

Now our three-minute recording of a real local run: deployment, an escrow purchase, the licence and the payout, and then the test results. The video shows 21 selected application tests and 10 EVM escrow tests. [Play video]

Slide 6 — Security Kalybek Zhumabekov · ~0:35

Thank you. Here are the controls that actually exist. Against reentrancy, we clear the credit before the transfer and use a nonReentrant guard. Against tampered or replayed quotes, we use EIP-712 signatures with buyer, expiry and quote ID checks. Against unauthorised settlement, every step checks the wallet and the state. Some risk remains: the trusted signer could authorise bad terms, and a compromised arbiter would affect disputes. We do not claim a multisig, an emergency pause or signer rotation. An independent audit is still planned.

Slide 7 — Market alternatives Kalybek Zhumabekov · ~0:25

Who else is in this space? Stock licensing services focus on content discovery and usage terms. Infringement services such as Pixsy find unauthorised use. NFT marketplaces focus on token trading. ArtGuard's question is different: does an escrow workflow with a wallet-bound licence, and no token, add useful assurance? These are reference categories, not our clients or partners.

Slide 8 — Main risks Kalybek Zhumabekov · ~0:40

These are the risks that matter for an investor. The two highest scores, 20 each, are weak demand and a funding shortfall. Our responses are to validate retention and willingness to pay, and to stage the funding. Next, with a score of 15 each, come signer or arbiter compromise and regulatory classification. Under the Law on Digital Assets, the Law on Personal Data and possibly the AIFC rules, a real commercial launch would need a legal review specific to this model. Finally, false authorship and delivery disputes need a clear evidence, takedown and appeal process.

Slide 9 — Revenue model Yevgeniy Tyo · ~0:30

Now the finances. Creators choose a plan. Free costs nothing but carries a 10% sales commission. Premium is 10 dollars a month with 5% commission. Business is 50 dollars a month with no commission. There is also a 5-dollar, seven-day promotion, but the base forecast includes no promotion revenue until demand is proven.

Slide 10 — Assumptions Yevgeniy Tyo · ~0:30

Our monthly targets remain 400 licence sales by Month 12, 1,300 by Month 24 and 2,500 by Month 36. The average licence is 30 dollars and the blended commission is 6.5%. Processing is 2.5% and planning FX is 447.94 tenge per dollar. We develop and operate the project ourselves without salaries. There are no assumed staff hires. Audit, legal review, marketing and hosting are still included.

Slide 11 — Three-year forecast Yevgeniy Tyo · ~0:30

Platform revenue grows from 6.50 million tenge in Year 1 to 71.39 million in Year 3. With cash payroll at zero, operating cash flow after planning tax is minus 5.75 million, then plus 15.81 and 37.99 million. The first-year loss comes from launch and security costs before sales grow. Unpaid founder time is a non-cash contribution; its opportunity cost is excluded.

Slide 12 — NPV Yevgeniy Tyo · ~0:30

At a 25% discount rate, three-year cash NPV is positive 24.97 million tenge. Monthly EBITDA first becomes positive in Month 8, and cumulative operating cash recovers earlier costs in Month 19. The optional terminal-growth case gives 127.10 million, but relies on a speculative perpetual-growth assumption. The positive cash result depends on unpaid founder work and unvalidated customer targets.

Slide 13 — Sensitivity Yevgeniy Tyo · ~0:30

We tested 30% fewer customers, 20% higher operating costs, and both together. Cash NPV falls to 11.74 million, 19.72 million and 6.83 million. Peak cash need ranges from 8.00 to 11.10 million, within the old 40-million proposal. All cases assume unpaid founders; paid staffing requires a separate scenario.

Slide 14 — Funding terms Assan Dairabaev · ~0:35

For comparison, we retain the earlier proposal: 25 million tenge in Month 1 and 15 million in Month 10 for 26% equity. But the corrected base peak cash need is only 7.52 million, so that raise should be resized. The illustrative exit gives 2.05 times invested capital and 29.9% annual IRR. These depend on an unverified exit multiple, customer growth and unpaid work. We recommend a staged validation pilot before agreeing funding terms.

Slide 15 — Team Assan Dairabaev · ~0:15

Our roles: Demezhan is CEO and led architecture and security; I am CFO and prepared the introduction and the pitch; Yevgeniy is CTO and built the financial model; Kalybek is COO and covered risk and regulation. Each of us documents our own commits.

Slide 16 — Conclusion Kalybek Zhumabekov · ~0:25

The local prototype demonstrates registration, escrow, a wallet-bound licence and a co-author withdrawal. With unpaid founder development, the corrected three-year cash NPV is positive 24.97 million tenge and cumulative cash recovery is Month 19. This excludes founder opportunity cost and does not prove demand. Our next steps are to validate customers and labour capacity, review security and resize the original funding proposal.

Slides 17–18 — AI disclosure and references Assan Dairabaev · ~0:15

Finally, we used Codex with GPT-6.1 Sol for substantial help with code, tests, the model and editing. Every team member has reviewed the work and can explain it. Our references are on the last slide and in the report. Thank you. We are happy to take your questions.